UNIT I

Introduction to the Development of Industrial Safety & Management

history & development | Factories Act | safety & productivity | safety organizations | committees | role of management & Govt.

I. Introduction - basic terms

Industrial safety is the branch of engineering/management that identifies the hazards in an industry, evaluates the risk and puts controls in place so that **workers, machines, materials and the environment** are protected from harm. A safe plant is also an efficient plant.

Term	Meaning (easy definition)
Hazard	Any source/situation with the potential to cause injury, illness, damage or loss (e.g. oil on floor, exposed wire).
Risk	Chance (probability) that a hazard actually causes harm x how severe the harm is. Risk = Probability x Severity
Accident	An unplanned, uncontrolled event that results in injury, ill-health or damage.
Incident	Any unwanted event that could / did cause harm; wider term than accident.
Near miss	Unplanned event that did NOT cause injury or damage but had the potential to - a free warning!
Unsafe act	Human behaviour that deviates from safe practice (no PPE, removing a guard, horseplay, shortcuts).
Unsafe condition	Physical defect in the workplace (no guard, poor lighting, wet floor, faulty tool).
Disabling injury	Injury that prevents the worker from doing normal work for one full day (shift) or more.

Objectives of industrial safety

> To **prevent injuries**, deaths and occupational diseases of workers.

> To protect plant, machinery, raw material and finished goods from damage / fire / explosion.

> To reduce the **cost of accidents** (medical, compensation, lost production, insurance premium).

> To comply with legal duties (Factories Act, rules) and avoid penalties and prosecution.

> To improve morale, goodwill and the public image of the company.

> To build a **safety culture** where every person feels responsible - 'safety is everybody's business'.

II. History & development of industrial safety

2.1 Early period

> **Ancient times**: Code of Hammurabi (c. 1750 BC) fixed penalties for builders whose faulty work killed people; Egyptians & Romans used simple protective measures in mines.

> **Before 1750**: work was done in homes / small workshops (cottage industry); hazards were limited and mostly individual.

> **Industrial Revolution (1760-1840, England)**: steam power, textile mills, coal mines. Unguarded belts & shafts, long hours (12-16 hrs), **child labour**, poor ventilation, no compensation - accidents and diseases rose sharply.

> Public protest and reports led to the first **factory laws**, and employers slowly realised that accidents cost money.

2.2 Milestones in the world

Year	Event / development
1802	UK **Health and Morals of Apprentices Act** - first law on working conditions in cotton mills (cleanliness, ventilation, limited hours).
1833	UK Factory Act - appointed the first **factory inspectors**; banned work by children under 9 in textile mills.
1844-1878	Further Factory Acts - machinery fencing, 10-hour day for women & young persons, consolidated in the Factory & Workshop Act 1878.
1893	Railway Safety Appliance Act (USA) - automatic couplers & air brakes on trains.
1911	Triangle Shirtwaist fire, New York (146 workers died) - locked exits; triggered fire-safety & factory laws.
1913	**National Safety Council** (USA) founded - first voluntary safety body.
1919	**ILO** (International Labour Organization) formed - international labour standards, incl. safety.
1931	H.W. **Heinrich** publishes 'Industrial Accident Prevention' - domino theory, ratio study.
1970	**OSHA Act** (USA) - employers must give a workplace free from recognised hazards. NIOSH created.

Year	Event / development
1974	UK **Health and Safety at Work Act (HSWA)** - goal-setting law, general duty of care, Safety Representatives.
1980s-90s	Bhopal (1984), Chernobyl (1986), Piper Alpha (1988) -> process safety, HAZOP, safety cases, SEVESO rules.
2018	**ISO 45001** - international standard for Occupational Health & Safety Management Systems (replaced OHSAS 18001).

2.3 Major industrial disasters - lessons learned

Disaster	Cause / lesson
Flixborough, UK (1974)	Temporary pipe bypass failed -> vapour cloud explosion, 28 dead. Lesson: **management of change**.
Seveso, Italy (1976)	Runaway reaction released dioxin. Led to the EU SEVESO Directive on major-accident hazards.
Bhopal, India (Dec 1984)	Methyl isocyanate (MIC) leak from Union Carbide plant; thousands died. Lesson: safety systems must work, emergency planning, community right-to-know. Led to **Factories Act amendment 1987** (Ch. IVA hazardous processes).
Piper Alpha, North Sea (1988)	Gas explosion on offshore platform, 167 dead. Lesson: permit-to-work and safety management.
Chasnala mine, India (1975)	Water inrush into coal mine, 375 miners died. Lesson: mine safety, flood protection.
Vizag gas leak, India (2020)	Styrene vapour leak from a polymer plant after lock-down restart; 12 died. Lesson: safe restart & storage of monomers.

2.4 Development of industrial safety legislation in India

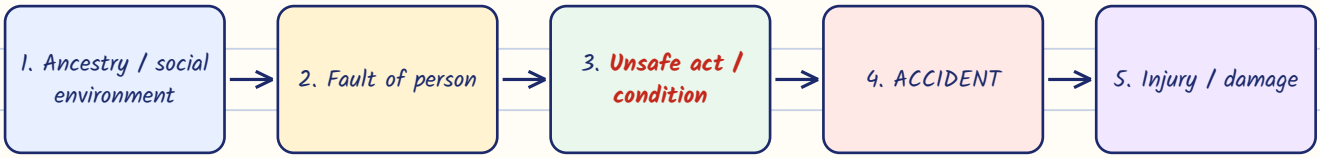
Year	Development
1881	First Indian **Factories Act** - child labour (7-12 yrs) restricted; machinery fencing.
1891, 1911, 1922	Revised Factories Acts - hours of work, women's hours, wider definition of factory.
1923	Workmen's Compensation Act and Indian Boilers Act.
1929-31	**Royal Commission on Labour** (Whitley Commission) - recommended safety measures and inspectors.

Year	Development
1934	Factories Act - 54-hr week, strong inspection system.
1948	**Factories Act, 1948** (in force 1 April 1949) - replaced the 1934 Act; also the ESI Act 1948.
1976	Factories Act amended - extended to wider categories, workers' participation.
1987	Post-Bhopal amendment - Chapter IVA: hazardous processes, site appraisal committee, safety officers, disclosure of information, workers' right to warn.
2020 / 2025	Parliament passed the **OSH Code 2020**; brought into force on 21 Nov 2025.

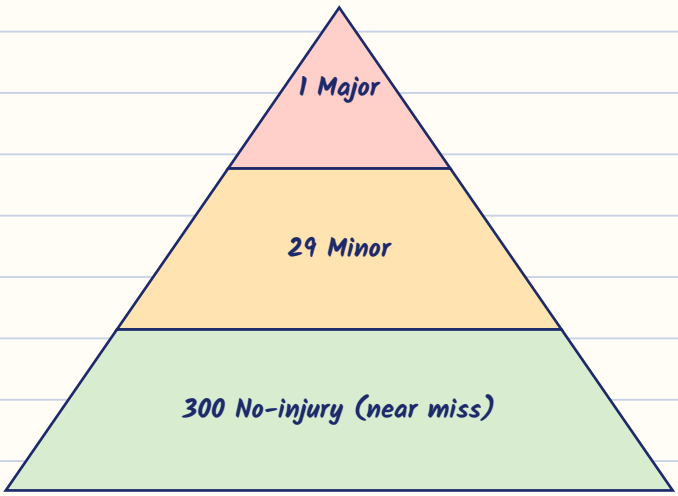
III. Theories of accident causation (basis of safety)

3.1 Heinrich's Domino theory (1931)

Accident is a chain of five dominoes. If any one domino is removed the chain stops. The **key domino** to remove is the **3rd - unsafe act / unsafe condition**.



3.2 Heinrich's accident ratio and Bird's ratio



Heinrich 1 : 29 : 300

1 major injury

29 minor injuries

300 no-injury events

Bird 1 : 10 : 30 : 600

serious : minor : damage : near-miss

88% unsafe acts, 10% unsafe conditions, 2% unavoidable

> Message: **reduce the base** (unsafe acts & near-misses) and the top of the pyramid - the serious injuries - will fall by itself.

> Other theories: **Multiple-causation theory**, **Accident-proneness theory**, **Energy-release theory**

(Haddon), **Human-factor / Ferrell theory**, **Systems & Fault-tree approach**.

IV. Implementation of the Factories Act, 1948

The **Factories Act, 1948** is the main central law in India that regulates **safety, health, welfare, working hours and employment of young persons** in factories. It is administered by the **State Governments** through the **Chief Inspector of Factories** and the **Directorate General Factory Advice Service & Labour Institutes (DGFASLI)** at Centre.

4.1 Important definitions (Sec. 2)

Term	Meaning under the Act
Factory [2(m)]	Premises where **10 or more** workers work using power, or **20 or more** workers working without power, on any day of the preceding 12 months, in a manufacturing process.
Worker [2(l)]	Person employed (directly / through agency, with or without wages) in a manufacturing process or related work; excludes members of armed forces.
Occupier [2(n)]	Person who has ultimate control over the affairs of the factory (director / partner in a company). He carries the **main legal responsibility**.
Manufacturing process	Making, altering, repairing, packing, washing, cleaning, treating or adapting any article, generating power, etc. [2(k)].
Hazardous process	Process listed in the First Schedule (e.g. chemicals, petroleum, metal extraction, fertilisers) where raw material/product may cause impairment of health or pollution.
Adult / Adolescent / Child	Adult >= 18 yrs; Adolescent 15-18 yrs; Child < 15 yrs (employment of children below 14 is prohibited).

4.2 Structure of the Act - chapter-wise scheme

Chapter (Sections)	Subject	Key provisions
II (8-10)	Inspecting staff	Inspectors appointed by the State; powers to enter, inspect, take samples, seize records, order measures.
III (11-20)	**Health**	Cleanliness, waste disposal, ventilation & temperature, dust/fume, overcrowding (14.2 m3 per worker), lighting, drinking water, latrines, spittoons.

Chapter (Sections)	Subject	Key provisions
IV (21-41)	**Safety**	Fencing of machinery (21), hoists & lifts (28), lifting machines (29), pressure plant (31), floors/stairs (32), protection of eyes (35), fire precautions (38), safety of buildings (40).
IVA (41A-41H)	Hazardous processes	Site appraisal committee, compulsory disclosure, permissible exposure limits, on-site emergency plan, workers' right to warn.
V (42-50)	**Welfare**	Washing, clothing, sitting, first-aid box, canteen, shelters, creches, welfare officers.
VI (51-66)	Working hours of adults	48 hrs/week, 9 hrs/day, rest interval, weekly holiday, overtime at double rate.
VII (67-77)	Young persons	Child < 14 not allowed; adolescents need fitness certificate; limited hours (4.5 hrs/day).
VIII (78-84)	Annual leave with wages	1 day for every 20 days worked (adult).
IX-XI	Penalties & procedure	Fines/imprisonment; general duties of occupier (7A) and manufacturers (7B).

4.3 Key safety provisions in simple words

> **Sec 21 – Fencing of machinery**: every dangerous part (moving parts, gears, belts, flywheels) must be securely fenced while in motion.

> **Sec 28 / 29**: hoists, lifts, cranes, chains, ropes and lifting tackle must be of good construction, tested and examined by a competent person (every 6 / 12 months) and marked with Safe Working Load (SWL).

> **Sec 31**: pressure plant (boilers, air receivers) must be operated within safe pressure; safety valves and examination required.

> **Sec 35**: eye protection (goggles / screens) for processes that produce flying particles or harmful radiation.

> **Sec 38**: means of escape, fire alarms, extinguishers, doors opening outwards, free passage to exits; workers trained in fire drill.

> **Sec 7A / 7B**: occupier must ensure health, safety & welfare of all workers; manufacturers/designers must test and give information on safe use of articles.

4.4 How the Act is implemented (mechanism)

1. **Registration & licensing**: the occupier gives notice, gets plans approved and a **licence** from the Chief Inspector before starting; renews it yearly.
2. **Inspection**: Inspectors visit factories, examine machines, records, registers (muster roll, accident register) and issue improvement notices / prohibitory orders.
3. **Reporting**: accidents causing death or disablement for 48 hrs or more must be reported (Form 18) - dangerous occurrences & occupational diseases too.
4. **Safety staff**: safety officers (Sec 40B), welfare officers (Sec 49), safety committees (Sec 41G) in notified factories.
5. **Enforcement**: prosecution - fine up to Rs 2 lakh and/or imprisonment up to 2 yrs for contravention; higher if death/ injury results.

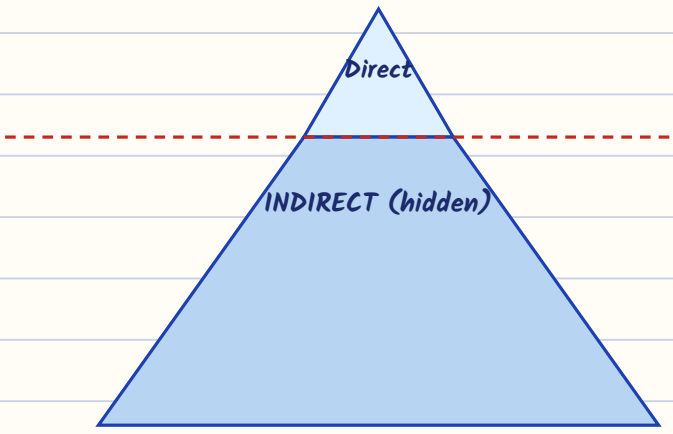
Latest update (exam tip)

> The **Factories Act, 1948** has now been subsumed by the **OSH Code, 2020** (Occupational Safety, Health & Working Conditions Code), in force from **21 Nov 2025**; the ESI Act and Employees' Compensation Act went into the Code on Social Security.

> Most safety duties (fencing, hazardous process, welfare, hours) are carried forward - JNTU/university syllabus still teaches the **Factories Act, 1948** so answer in those terms.

V. Safety and productivity

Many managers think safety **slows production**. Experience shows the opposite - **safe plants are productive plants**.



Direct (visible): medical, compensation, repair of damage - about **1 part**

Indirect (hidden): lost time, damaged plant, retraining, lower morale, legal - about **4 parts** (Heinrich)

Safety improves productivity by...	Explanation
Fewer lost hours	Less time lost in accidents, investigations, replacement of injured worker.
Better machine condition	Guarding, maintenance & housekeeping reduce breakdowns and rework.
Higher morale	Workers who feel cared for work with confidence and loyalty; lower absenteeism and labour turnover.
Lower cost	Lower insurance premium, compensation, medical bills; fewer damaged goods and penalties.
Quality & image	Orderly work-place gives better quality; good reputation attracts customers and talent.

Safety + Productivity + Quality = Profit ('**Safety is good business**')

> **Productivity** = Output / Input. Accidents reduce output and increase input (cost), so they directly reduce productivity.

> Measure of safety performance: **Frequency Rate**, **Severity Rate**, **Incidence Rate** (see formulae in Unit II).

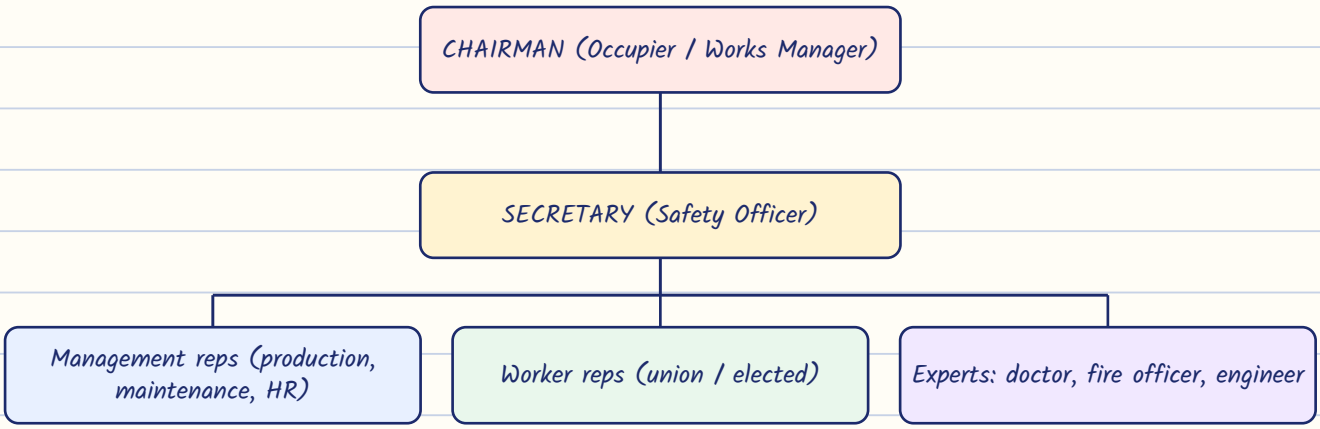
VI. Safety organizations

Organization	Level	Role / contribution
ILO	World	UN agency (1919, Geneva); sets conventions & codes (e.g. CISS OSH Convention); training, research.
OSHA / NIOSH	USA	OSHA makes & enforces standards; NIOSH does research & recommends limits.
BSC / RoSPA / HSE	UK	British Safety Council, Royal Society for Prevention of Accidents, Health & Safety Executive - training, awards, regulation.
NSC (India)	India	National Safety Council (1966, Mumbai) - promotes safety awareness, training, safety awards, National Safety Day (4 March).
DGFASLI	India	Under Ministry of Labour; advises Govt, runs Central Labour Institute (Mumbai) & Regional Labour Institutes; training, surveys.
BIS	India	Bureau of Indian Standards - IS codes for PPE, extinguishers, safety signs, machine guarding.

Organization	Level	Role / contribution
PESO	India	Petroleum & Explosives Safety Organisation - licences for explosives, petroleum, gas cylinders.
AERB	India	Atomic Energy Regulatory Board - radiation safety.
NFPA / ISO / ANSI	Intl.	Codes & standards - NFPA (fire), ISO 45001 (OHS management), ANSI (safety standards).

VII. Safety committees and structure

A **Safety Committee** is a joint body of management and workers that meets regularly to discuss safety problems and suggest remedies. Under **Sec 41G** (Factories Act) the occupier of a factory carrying on a hazardous process must set up a safety committee with **equal representation** of workers and management.



Functions of the safety committee

> Study accident reports, find causes and recommend **preventive action**.

> Conduct **safety inspections / tours** of shop floors, review hazards and near-misses.

> Suggest improvements in **work methods, PPE, housekeeping, training**.

> Organise **safety campaigns**, posters, competitions, safety week & mock drills.

> Review progress of previous recommendations, keep minutes & follow-up record.

> Act as a **two-way channel** between workers and management - build trust and participation.

Working of the committee

Point	Detail
Composition	Chairman (occupier/manager), secretary (safety officer), equal management & worker reps; doctor / fire officer as members.
Meetings	At least once a month (agenda circulated in advance); special meeting after a major accident.
Minutes	Recorded, signed, sent to departments; action points with a responsible person and a target date.
Types	Plant (central) committee, department / shop committees, contractor-safety committee.

VIII. Role of management in industrial safety

> **Top management commitment**: written **safety policy** signed by the CEO, displayed and followed - 'safety first, production next' is shown by actions, not slogans.

> **Organisation**: line managers are responsible for safety of their people; a **safety officer** (staff function) advises, trains, audits.

> **Resources**: budget for guards, PPE, training, safe layout, maintenance and emergency equipment.

> **Planning**: hazard identification & risk assessment (HIRA), safe work procedures, permit-to-work, emergency preparedness.

> **Training & motivation**: induction & refresher training, rewards for safe work, involve workers in decisions.

> **Monitoring**: inspections, audits, accident investigation, statistics, corrective action - 'plan-do-check-act'.

> **Legal compliance**: meet all statutory requirements; keep registers; cooperate with inspectors.

Line vs Staff in safety

> **Line**: supervisors & managers have authority and are **accountable** for safe work in their area.

> **Staff**: safety department **advises** (no line authority) - training, inspections, statistics, investigations.

> Under law a **Safety Officer** is required in factories with 1000 or more workers, or those with hazardous processes (Sec 40B).

IX. Role of Government in industrial safety

> **Legislation**: makes laws & rules - Factories Act, Boilers Act, Explosives Act, ESI, compensation, OSH Code.

> **Enforcement**: Factory Inspectorates, Boiler inspectorate, PESO inspect, issue notices, prosecute violators.

> **Standards & research**: BIS standards, DGFASLI / CLI studies, permissible exposure limits.

> **Training & education**: Labour Institutes, ESI hospitals, safety awards (Vishwakarma Rashtriya Puraskar), National Safety Week.

> **Social security**: compensation and ESI medical care, disability & dependant benefits.

> **Emergency preparedness**: Disaster management (NDMA), on-site/off-site emergency plans for MAH units, Public Liability Insurance Act 1991.

> **International role**: ratifies ILO conventions and coordinates with global agencies.

IX-A. Safety policy, safety officer & 3 E's

Safety policy statement

A good policy is short, written, signed by top management, displayed on notice boards and reviewed yearly. It should state: (1) the **commitment** to provide a safe & healthy workplace, (2) **responsibilities** of managers, supervisors, workers, (3) the duty to **report** hazards, (4) **compliance** with law and (5) **continual improvement**.

Sample policy lines

> 'No job is so important that it cannot be done safely.'

> 'Every employee has the right to refuse unsafe work and the duty to report it.'

> 'Safety performance will be part of every manager's appraisal.'

Duties of a safety officer (Sec 40B, Factories Rules)

> Advise management on safety laws and **safe procedures**; assist in **accident investigation**.

> Inspect the plant regularly; check guards, PPE, fire equipment, housekeeping.

> Maintain **accident records**, compute frequency & severity rates and report to the Inspector.

> Train workers (induction, tool-box talks, mock drills) and run safety weeks / campaigns.

> Act as **secretary** of the safety committee and follow up recommendations.

> Qualification: a recognised degree/diploma + a diploma in industrial safety; minimum age and experience as prescribed by the State rules.

The 3 E's of safety

E	Meaning	Examples
Engineering	Make the machine / process safe by design	Guards, interlocks, ventilation, safe layout, isolation
Education	Make the worker knowledgeable and willing	Training, posters, safety talks, demonstrations
Enforcement	Make sure rules are followed	Discipline, supervision, audits, penalties, rewards

Principles of accident prevention

1. Fix **responsibility** - line management owns safety.
2. Find **root causes** of accidents and near-misses, not just the victim's fault.
3. **Eliminate** the hazard first; use PPE only as the last line of defence (hierarchy of control).
4. Select, train and place the **right person on the right job**.
5. Keep records, analyse statistics and use them to **plan action**.
6. Involve workers - their suggestions are the cheapest hazard-detection system.

> Hierarchy of control: **Eliminate** > Substitute > Engineering > Administrative > PPE

Safety performance indices (used by all safety bodies)

Index	Formula
Frequency rate (FR)	FR = (No. of disabling injuries x 10^6) / Man-hours worked
Severity rate (SR)	SR = (Man-days lost x 10^6) / Man-hours worked
Incidence rate (IR)	IR = (No. of injuries x 1000) / Average no. of workers employed
Avg. days lost per injury	= Severity rate / Frequency rate

Eg: 12 disabling injuries, 150 days lost, 2,000,000 man-hours -> FR = 12x10^6/2x10^6 = **6**; SR = 150x10^6/2x10^6 = **75**.

Welfare & hours of work - quick table (Factories Act)

Facility / rule	Requirement
First-aid box (Sec 45)	One for every 150 workers; trained person in charge; ambulance room if 500+ workers
Canteen (Sec 46)	Compulsory if 250+ workers
Rest room / shelter (Sec 47)	If 150+ workers

Facility / rule	Requirement
Creche (Sec 48)	If 30+ women workers; for children below 6 yrs
Welfare officer (Sec 49)	If 500+ workers
Weekly hours (Sec 51)	48 hrs/week; daily 9 hrs (Sec 54); spread-over 10.5 hrs (Sec 56)
Rest interval (Sec 55)	At least 30 min after 5 hrs of continuous work
Overtime (Sec 59)	Twice the ordinary rate of wages

X. Quick revision & important questions

Remember (one-liners)

> Factory = 10+ workers with power / 20+ without power.

> Occupier carries the main responsibility under the Factories Act.

> Heinrich ratio 1:29:300; 88% accidents due to unsafe acts.

> Safety committee = management + workers (equal reps), meets monthly.

> Bhopal 1984 -> 1987 amendment (hazardous processes).

> NSC India founded 1966 - National Safety Day 4 March.

1. Trace the history and development of industrial safety in India. (10 M)
2. Explain the salient features of the Factories Act, 1948 regarding safety. (10 M)
3. Discuss the relation between safety and productivity. (5 M)
4. Describe the structure and functions of a safety committee. (10 M)
5. Explain the role of management and Government in industrial safety. (10 M)
6. Write short notes on safety organizations (ILO, NSC, DGFASLI). (5 M)